

Spatial Analysis of Older Adult Population and Hospital Accessibility in Alachua County, Florida

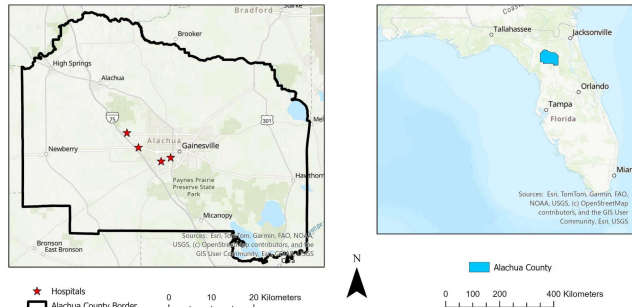
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GIS 3043 Foundations of Geographic Information

University of Florida, Department of Geography

Introduction

Alachua County Study Area Map



Population aging is becoming an increasingly serious issue in the United States and has become an important topic. Older adults typically have greater needs for social services, and an important one among these is healthcare. If elderly people can't access the necessary amenities, it could lead to serious illness such as vulnerability and dependence (Guida et al., 2022). In emergency situations, the distance between the hospital and their residence is important. Therefore, studying the geographical relationship between older adults and the hospital location is very important.

Studies show that GIS is well-suited for analyzing issues related to the spatial distribution of elderly communities and accessibility to amenities (Choi et al., 2021). This project examines the relationship between the distribution of the elder adults population and the hospitals within Alachua County, including whether there is a mismatch and whether it is clustered. This can help local government to prioritise this problem and improve in the future.

Research Questions

How are older adult populations distributed across Alachua county?

How does hospital accessibility differ across Alachua county?

Are low access older adults in Alachua county clustered?

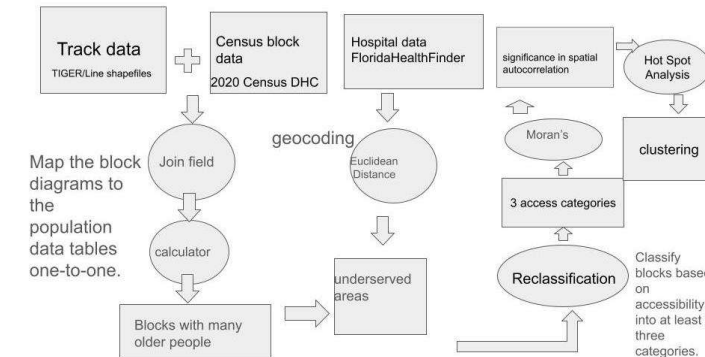
Materials & Methods

1. Data sources

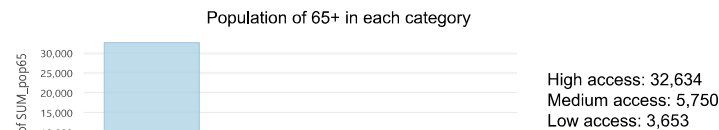
- U.S. Census Bureau TIGER/Line shapefiles (2020 census block boundaries)
- U.S. Census Bureau 2020 Census DHC
- FloridaHealthFinder hospital information

2. Main Tools and Methods

- Geocoding of hospital addresses
- Euclidean Distance for hospital accessibility
- Reclassification of blocks into access categories
- Summary statistics by access category
- Moran's I for global spatial autocorrelation and clustering
- Hot Spot Analysis for clustering



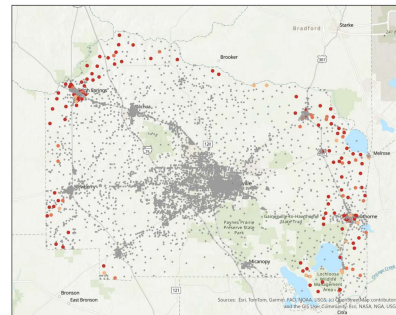
Results



Moran's I Result

Moran's Index: 0.178905
z-score: 60.451848
p-value: 0.000000

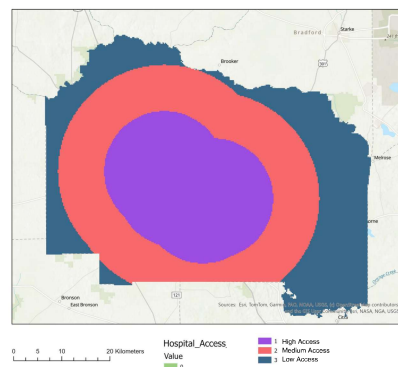
Hot Spot Map of Risk Population



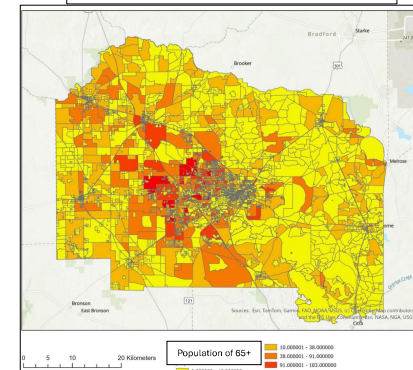
Main findings

- Most older adults in Alachua County live in high hospital access regions.
- Moran's I's result shows that the low hospital access elderly population is spatially clustered.
- Hot Spot Analysis shows that low hospital access elderly population is clustered near the county boundaries.

Hospital Access Category Map



Block-level Older Adult Population Map



Discussion

General Findings and interpretation

Most of the elderly adults in Alachua county live in area with high hospital accessibility. Generally speaking, the area near county center, like the city of Gainesville, has high hospital accessibility. However, there are a lot of hot spots at the edge of the county, showing that the unevenness of hospital access still exist.

Limitations

- Euclidean distance is not the real estimated travel time on the road. Cause road network is also uneven, the travel time from each block to the hospital require further research.
- Other kinds of small healthcare services like clinics or pharmacies are not included in the research
- The research only includes hospitals inside Alachua county boundaries. So the accessibility for the blocks near the boundaries can be underestimated.

Future Research

- Use the road data to estimate the travel time then classify the accessibility with it.
- Include other healthcare services like clinics or pharmacies and weight it differently
- Include the hospital or other health care services in neighboring counties.

Conclusion

The distribution of elderly adults and hospital accessibility is not evenly distributed in Alachua county. Even most elderly population live in high access areas, there are still many clusters of elderly who has low access to hospitals. The county can use the resource to help these blocks and communities.

References

- Choi, K., Lee, Y., & Basrak, Z. (2021). Identifying communities of concern for older adults using spatial analysis: Focusing on accessibility to health, social, and daily services. *Journal of Applied Gerontology*, 40(11), 1371-1381. <https://doi.org/10.1177/0733464820978000>
- Guida, C., Carpentieri, G., & Masoumi, H. E. (2022). Measuring spatial accessibility to urban services for older adults: An application to healthcare facilities in Milan. *European Transport Research Review*, 14, Article 23. <https://doi.org/10.1186/s12544-022-00544-3>